

When connecting two or more graphics cards together there are a few rules that need to be followed. NVIDIA requires that both cards be identical in series and model i.e., a 660Ti must be paired with another 660Ti. The brand need not be the same but to avoid any compatibility issues at all then stick to the same brand. Then you need a motherboard which supports SLI / CrossFire configuration and an SLI bridge which is what connects the two cards together. AMD on the other hand requires the family to be the same but there is no need for the model to be the same. So a 7770 can be paired with a 7970 without any issues. Also a motherboard with CrossFire / SLI support and a crossfire bridge are the other components. If at all you can't find two NVIDIA cards that are the same then you can still hook the two cards together in SLI except one of the cards will become a dedicated PhysX card, we'll get to this in a bit.

PCIE 2 / 3

A graphics card connects to the motherboard via the PCIE slot because of the high bandwidth requirement of a single card.

That too it must be a x16 PCIE slot. However, when you go for an SLI / CrossFire configuration

you need not need x16 bandwidth

on the individual slots. Tests have

proven that there is only a miniscule hit in performance when two x8 slots are used instead of two x16 slots. This test was run on a PCIE 2.x generation slot, so it still remains true for a PCIE 3.0 slot.



PhysX/Havok

PhysX and Havok are two physics APIs. Game developers choose either software API to handle the in-game physics like smoke effects, character model ragdoll effects and general environment interaction. Havok is supported on both NVIDIA and AMD cards while PhysX is purely supported on NVIDIA cards since NVIDIA bought their parent company.